

Simplified Schema Table

Trace Type	Specificity	Invocation	Duration	Memory
Azure2019	Per-function	Count (at each minute)	Percentiles of average execution time	Percentiles of average allocated memory
Huawei2023	Per-function	Count (at each second)	Avg function execution time (at each minute)	Average usage over all pods (at each minute)
Azure2021	Per-invocation	Timestamp	Seconds	N/A {Empirical 200MB chosen}
IBM2026	Per-invocation	Timestamp	Milliseconds	N/A {Empirical 200MB chosen}

Conceptually:

Traces fall into 2 main formats:

- Per-function
 - Aggregated statistics of a function's invocations
 - Azure2019 & Huawei2023
- Per-Invocation
 - Timestamp and duration for each invocation
 - Azure2021 & IBM2026

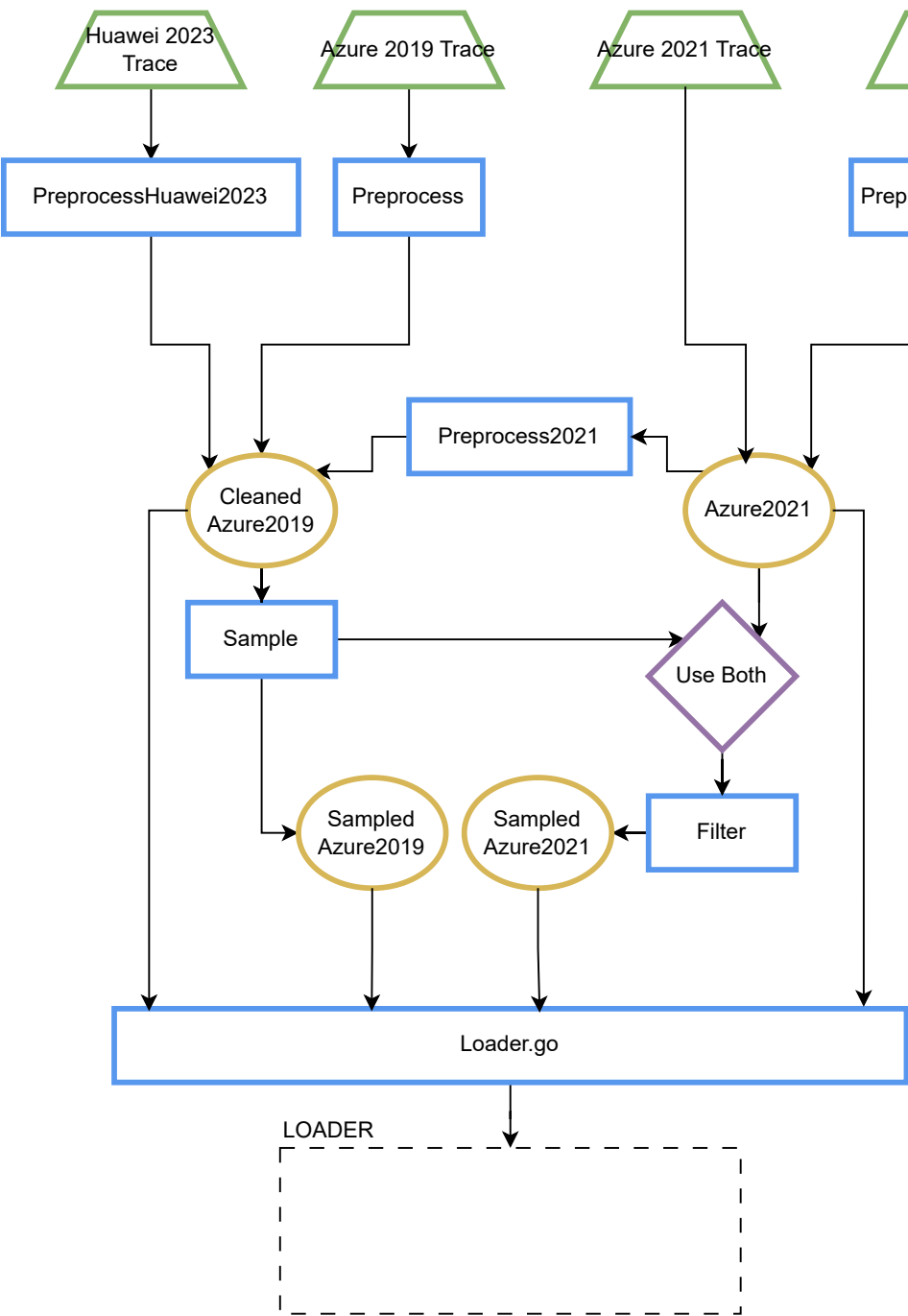
Loader and Sampler natively accept traces in the 2 main formats:

- Per-function (Azure2019)
- Per-invocation (Azure2021)

Support for additional traces is achieved by transforming the trace into either Azure2019 or Azure2021 format, whichever format is more similar.

- Azure2019 format
 - Huawei2023
- Azure2021 format
 - IBM2026

Main Process Flowchart



Legend	
	Original Trace Data
	Process / Conversion
	Internal Trace Format
	Multiple Inputs

Comparison Table

Trace Type	Intermediate Format	General Conversion	Invocation	Duration	Memory
Azure2019	Per-function (Cleaned Azure2019)	Time-interval filtered. Drop functions with incomplete info.	Sample within time-unit (Invocation count per minute) using user-indicated distribution type.	Randomly sampled from described distribution quartiles.	Randomly sampled from described distribution quartiles. Described memory for application divided evenly among its functions.
Huawei2023	Per-function (Cleaned Azure2019)	Time-interval filtered. Drop functions with incomplete info.	Sample within time-unit (Invocation count per minute) using user-indicated distribution type.	Distribution quartiles calculated from non-zero data points (execution time averaged over all pods, per minute). Randomly sampled from distribution quartiles.	Distribution quartiles calculated from non-zero data points (memory allocated across all pods, per minute). Randomly sampled from distribution quartiles.
Azure2021	Per-invocation (Azure2021)	Time-interval filtered. Removed functions with 0ms invocations above threshold percent.	Calculated `start_timestamp` from `end_timestamp` and `duration`.	Directly used	Surrogate value of 200MB (derived from Azure2019)
IBM2026	Per-invocation (Azure2021)	Time-interval filtered. Treat `AppHash` as individual function, `NamespaceHash` as application	Timestamp zeroed to start of time-interval.	Directly used	Surrogate value of 200MB (derived from Azure2019)

METHODOLOGY:

Comparsion on how an input trace data type, becomes interpreted at loader

Function Invocation Counts

- 14 files, one file per 24-h period: `invocations_per_function_md.anon.d[01-14].csv`

Schema	
Field	Description
HashOwner	unique id of the application owner ¹
HashApp	unique id for application name ¹
HashFunction	unique id for the function name within the app ¹
Trigger	trigger for the function ²
1 .. 1440	1440 fields, with the number of invocations of the function per each minute of the 24h period in the file ³

Function Execution Duration

14 files, one file per 24-h period: `function_durations_percentiles.anon.d[01-14].csv`

Schema	
Field	Description
HashOwner	unique id of the application owner
HashApp	unique id for application name
HashFunction	unique id for the function name within the app
Average	Average execution time (ms) across all invocations of the 24-period ⁴
Count	Number of executions used in computing the average ⁵
Minimum	Minimum execution time for the 24-hour period ⁶
Maximum	Maximum execution time for the 24-hour period ⁶
percentile_Average_0	Weighted 0th-percentile of the execution time <i>average</i> ⁷
percentile_Average_1	Weighted 1st-percentile of the execution time <i>average</i> ⁷
percentile_Average_25	Weighted 25th-percentile of the execution time <i>average</i> ⁷
percentile_Average_50	Weighted 50th-percentile of the execution time <i>average</i> ⁷
percentile_Average_75	Weighted 75th-percentile of the execution time <i>average</i> ⁷
percentile_Average_99	Weighted 99th-percentile of the execution time <i>average</i> ⁷
percentile_Average_100	Weighted 100th-percentile of the execution time <i>average</i> ⁷

Sample of "Invocation Counts" .csv data						
HashOwner	HashApp	HashFunction	Trigger	1	2	3
71ca12c7af70d021e285b51b245942f8432df6463ff9f2c009b06b3f661f871f	734272c01926d19690e5ec308bab64ef97950b75b1c7582283e0783fce1751d8	313c03f53a0d31f70aec25f62efb33e7dd779725ca4af579018452d1204beaad	http	1	0	1
dca6b75d51b494146b50f93b0f1a863ce99164942695e8309c5d64b47709064e	17c37a0fdd5d1932b755c0e6447137bc08fd524f455e14fdac414f584de08dc5	c9f8e30e36d1aef62c10b3cfca6e289a93848a148d876dd514753040314f4817	timer	52	47	50

Sample of "Execution Duration" .csv data					
HashOwner	HashApp	HashFunction	Average	Count	Minimum
5640c1597ef75fa9a7e9c6925022a039a4ba982924170953a51cfe4ef3a54ee9	5126901eff078c9a1f5295c859c9327588284a43cc2c0b4b261426b009ddfe6e	c968871b4ef0123401975d026b85cae2ad7dad9d06ae94f13b978bd66dfd063f	100	2880	0.0
5640c1597ef75fa9a7e9c6925022a039a4ba982924170953a51cfe4ef3a54ee9	5126901eff078c9a1f5295c859c9327588284a43cc2c0b4b261426b009ddfe6e	1bc2d86badd21b18a8533d8e961e52585e5ad1fee0e2c2885220a8463a39492d	202	365	0.0

Sample of "Execution Duration" .csv data (cont.)							
Maximum	percentile_Average_0	percentile_Average_1	percentile_Average_25	percentile_Average_50	percentile_Average_75	percentile_Average_99	percentile_Average_100
2595.0	0	0	0	1	1	1376	2595
2596.0	0	0	1	1	6	2326	2596

Application Memory

12 files, one file per 24-h period (last 2 are missing): `app_memory_percentiles.anon.d[01..12].csv`

Schema	
Field	Description
HashOwner	unique id of the application owner
HashApp	unique id for application name
SampleCount	Number of samples used for computing the average
AverageAllocatedMb	Average allocated memory across all SampleCount measurements throughout this 24h period ⁸
AverageAllocatedMb_pct1	1st percentile of the average allocated memory ⁹
AverageAllocatedMb_pct5	5th percentile of the average allocated memory ⁹
AverageAllocatedMb_pct25	25th percentile of the average allocated memory ⁹
AverageAllocatedMb_pct50	50th percentile of the average allocated memory ⁹
AverageAllocatedMb_pct75	75th percentile of the average allocated memory ⁹
AverageAllocatedMb_pct95	95th percentile of the average allocated memory ⁹
AverageAllocatedMb_pct99	99th percentile of the average allocated memory ⁹
AverageAllocatedMb_pct100	100th percentile of the average allocated memory ⁹

Sample of "Application Memory" .csv data											
HashOwner	HashApp	SampleCount	AverageAllocatedMb	AverageAllocatedMb_pct1	AverageAllocatedMb_pct5	AverageAllocatedMb_pct25	AverageAllocatedMb_pct50	AverageAllocatedMb_pct75	AverageAllocatedMb_pct95	AverageAllocatedMb_pct99	AverageAllocatedMb_pct100
71ca12c7af70d021e285b51b245942f8432df6463ff9f2c009b06b3f661f871f	7ca324d9fc836a5d4562811c11ce3719530ee919dd1fb91bcaf71942eab8240a	17235	108	108	108	108	109	109	109	110	110
5640c1597ef75fa9a7e9c6925022a039a4ba982924170953a51cfe4ef3a54ee9	5126901eff078c9a1f5295c859c9327588284a43cc2c0b4b261426b009ddfe6e	536284	322	167	199	251	307	357	532	692	991

Cleaned Azure2019

Very similar to Azure2021 with the following differences:

Invocation -> Instead of minute counts from 1 to 1440, only the relevant minute counts columns are kept.

Duration -> Exactly the same

Memory -> Each memory percentile data was at application level. Evenly splits them into equal function level percentile data.

Fields in invocations.csv
HashOwner
HashApp
HashFunction
Trigger
540
541
542
543
...

Fields in memory.csv
HashFunction
HashOwner
HashApp
SampleCount
AverageAllocatedMb
AverageAllocatedMb_pct1
AverageAllocatedMb_pct5
AverageAllocatedMb_pct25
AverageAllocatedMb_pct50
AverageAllocatedMb_pct75
AverageAllocatedMb_pct95
AverageAllocatedMb_pct99
AverageAllocatedMb_pct100

Azure2021

Schema

- app: application id (encrypted)
- func: function id (encrypted), and unique only within an application
- end_timestamp: function invocation end timestamp (in seconds)
- duration: duration of function invocation (in seconds)

[Azure2021](#)
[Original Repo](#)

Sample of .csv data

app	func	end_timestamp	duration
734272c01926d196 90e5ec308bab64ef 97950b75b1c75822 83e0783fce1751d8	313c03f53a0d31f7 0aec25f62efb33e7 dd779725ca4af579 018452d1204beaad	5160.142570018768	0.134
17c37a0fdd5d1932 b755c0e6447137bc 08fd524f455e14fd ac414f584de08dc5	c9f8e30e36d1aef6 2c10b3cfca6e289a 93848a148d876dd5 14753040314f4817	5161.280997037888	0.013

Each function invocation (app, func) ->
end_timestamp and duration

Huawei Private

This dataset contains 141 days (collected over 235 days) for 200 functions from all availability zones combined of our private cloud.

Metric	Minute	Second	Description
Function ID	-	-	Unique function identifier out of 200 (0-199)
Timestamp	-	-	Timestamp in seconds (0-20303940)
Requests	Requests per minute	Requests per second	Number of function invocations
Function delay	Function delay per minute	Function delay per second	Function execution time averaged over all pods running that function
Platform delay	Platform delay per minute	Platform delay per second	Platform delay is scheduling time and some network overheads; averaged over all pods running that function
CPU usage	CPU usage per minute	N/A	Percentage of allocated CPU used per function averaged over all pods
Memory usage	Memory usage per minute	N/A	Percentage of allocated memory used per function averaged over all pods
CPU limit	CPU limit per minute	N/A	Allocated CPU for all pods running that function (normalized)
Memory limit	Memory limit per minute	N/A	Allocated memory for all pods running that function (MB)
Instances	Instances per minute	N/A	Number of pods allocated to that function

Sample of .csv data

day	time	0	1
0	0	20	4
0	60	18	4
0	120	23	6

Each column -> unique function
`time` -> timestamp/seconds since 0 day 0
Each CSV is 1 day, days not continuous

▼ memory_limit_minute

day_000.csv

day_001.csv

day_002.csv

day_003.csv

day_004.csv

Information Included

- Hashed Namespace
- Hashed Application Name
- Daily Application Configurations (see `app_configs.pickle`)
 - 'NamespaceHash'
 - 'AppHash'
 - 'AppContainerRequestCpu' (in vCPU)
 - 'AppContainerRequestMemory' (in GB)
 - 'AppContainerConcurrency'
 - 'AppMinScale': The minimum number of instances for the application. If set to 0 (default value), the application scales down to zero.
- Weekly Traffic Data (10 weeks, see `week_*.pickle`)
 - 'NamespaceHash'
 - 'AppHash'
 - 'NumEvents'
 - 'InvocationTimes': A list of size `NumEvents` containing the invocation timestamps (in seconds, with millisecond-level accuracy)
 - 'AppExecTimes': A list of size `NumEvents` containing the application execution times (in milliseconds)
 - 'TotalExecTimes': A list of size `NumEvents` containing the total execution times (in milliseconds)
 - 'PodHash': A list of size `NumEvents` containing the pod hashes. Each app can have multiple pods, so this column allows to identify which pod served which request. Note that the ordering of requests across the these four lists is consistent.

```
df_week1.head()
```

Python

	NamespaceHash		AppHash	NumEvents	InvocationTimes	AppExecTimes	TotalExecTimes	PodHash
0	ce74bb5e28c68fa4b8b40fe36f17bda7	698c220abc1f1e2c0b1f2f0822f0b2cb		16	[452097.686, 452098.803, 452194.977, 452197.89...	[660.0, 355.0, 1936.0, 176.0, 2314.0, 4605.0, ...	[662.0, 357.0, 1938.0, 177.0, 2317.0, 4607.0, ...	[9e807f5022908bb7, 9e807f5022908bb7, 9e807f502...
1	ce74bb5e28c68fa4b8b40fe36f17bda7	3ef1e4394f86e641b4b7d858af1077da		5041	[18001.002, 18120.42, 18240.3, 18360.165, 1848...	[331, 166, 361, 232, 246, 223, 285, 274, 266, ...	[334, 168, 362, 234, 253, 227, 287, 276, 269, ...	[f503970b13341123, f503970b13341123, f503970b1...
2	ce74bb5e28c68fa4b8b40fe36f17bda7	815753a568994dbd89543508dfe353f3		92116	[18005.955, 18015.399, 18015.402, 18015.407, 1...	[67, 57, 32, 48, 33, 16, 23, 35, 31, 16, 22, 1...	[69, 59, 33, 50, 35, 17, 25, 36, 32, 18, 24, 1...	[eb2ae13eb0b9e43e, a35e144897ec65f6, 4b4357cbe...
3	ce74bb5e28c68fa4b8b40fe36f17bda7	90ca9ec59ea25f05c97334a360683352		67827	[18015.232, 18015.409, 18015.508, 18015.595, 1...	[24, 43, 28, 17, 20, 28, 57, 13, 25, 28, 32, 5...	[26, 44, 29, 18, 22, 29, 59, 19, 27, 29, 34, 5...	[824aee004e59ea05, 824aee004e59ea05, e08aeb560...
4	5ea634b02bc9a17a3aa86661f6fb8016	b14ac4041d74640fa87a6de6c093a3b4		1259	[29619.455, 29620.441, 29621.448, 29622.483, 2...	[7, 2, 7, 2, 2, 15, 10, 10, 2, 8, 2, 8, 2, 2, ...	[9, 4, 9, 4, 3, 16, 12, 12, 3, 10, 3, 10, 3, 4...	[16c4754cd1f867cf, 16c4754cd1f867cf, 16c4754cd...